

Learning Roadmap: Web development

Level: Beginner | Duration: 1 Months

Overview

This roadmap provides a comprehensive introduction to web development, covering the basics of HTML, CSS, JavaScript, and web development frameworks.

Phase 1: Introduction to Web Development

Duration: 1 weeks

Learn the basics of web development, including HTML, CSS, and JavaScript. Understand how to structure and style web pages, and introduce basic interactivity.

Topics & Resources:

- HTML and CSS

- * W3Schools HTML Tutorial
- * W3Schools CSS Tutorial

Project: Create a simple web page using HTML and CSS

- Introduction to JavaScript

- * W3Schools JavaScript Tutorial
- * Codecademy JavaScript Course

Project: Add basic interactivity to a web page using JavaScript

Milestones:

- [] Create a simple web page
- [] Understand basic JavaScript concepts

Phase 2: Front-end Development

Duration: 1 weeks

Dive deeper into front-end development, learning about responsive design, CSS frameworks, and front-end build tools.

Topics & Resources:

- Responsive Design

- * W3Schools Responsive Web Design Tutorial
- * CSS-Tricks Responsive Design Guide

Project: Create a responsive web page

- CSS Frameworks

- * Bootstrap Documentation
- * Material-UI Documentation

Project: Use a CSS framework to create a web page

Milestones:

- [] Create a responsive web page
- [] Understand how to use a CSS framework

Phase 3: Back-end Development Introduction

Duration: 1 weeks

Introduction to back-end development, covering the basics of server-side programming, databases, and back-end frameworks.

Topics & Resources:

- Introduction to Server-side Programming

- * W3Schools Server-side Programming Tutorial
- * MDN Web Docs Server-side Programming Guide

Project: Create a simple server-side application

- Introduction to Databases

- * W3Schools SQL Tutorial
- * MongoDB Documentation

Project: Create a simple database-driven application

Milestones:

- [] Understand basic server-side programming concepts
- [] Understand how to interact with a database

Phase 4: Project Development and Review

Duration: 1 weeks

Apply the skills learned throughout the month to a real-world project. Review and refine the project, ensuring it meets web development best practices.

Topics & Resources:

- Project Development

- * Project ideas on GitHub
- * Web development project tutorials on YouTube

Project: Create a comprehensive web development project

- Code Review and Refactoring

- * Code review guidelines on GitHub
- * Refactoring tutorials on Pluralsight

Project: Review and refactor the project code

Milestones:

- [] Create a comprehensive web development project
- [] Refactor the project code for best practices

Essential Resources

Recommended Courses

- FreeCodeCamp
- Udemy Web Development Course

Books

- HTML and CSS: Design and Build Websites
- JavaScript: The Definitive Guide

Communities

- Stack Overflow
- Reddit r/webdev

Tools

- Visual Studio Code
- Google Chrome Developer Tools